

Lakshmi Sahasra Jangoan

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PROFESSIONAL SUMMARY

Data professional with 2.5+ years of experience across data analytics, machine learning, and data engineering, turning large, messy, real-world datasets into forecasts, dashboards, and production-grade data systems. Skilled in SQL, Python, and BI tools (Power BI, Tableau) for exploratory analysis and stakeholder reporting, and in ETL pipeline design, statistical modeling (XGBoost, TensorFlow, PyTorch), and API containerized deployment (FastAPI, Docker) for scalable, validated data pipelines. Proven ability to frame ambiguous problems, catch data quality issues early, and communicate findings that drive measurable outcomes for technical and non-technical stakeholders alike.

TECHNICAL SKILLS

- Languages & Querying:** Python (Pandas, NumPy, Scikit-learn), SQL (CTEs, Window Functions, Joins, Subqueries, Aggregations), R, JavaScript, Bash
- Data Analysis & BI:** Power BI (DAX, Data Modeling), Tableau, Excel (Pivot Tables, VLOOKUP/XLOOKUP), Dashboard Development, KPI Reporting, Ad-hoc & Exploratory Data Analysis
- Machine Learning & Statistics:** Regression, Classification, XGBoost, Random Forest, TensorFlow, PyTorch, Time Series Forecasting (ARIMA), Hypothesis Testing, Cross-Validation
- Data Engineering & Tools:** ETL Pipeline Design, PostgreSQL, Azure SQL, FastAPI, Docker, RESTful APIs, Data Validation & Quality Assurance, Git, GitHub, Pytest, Matlab

PROFESSIONAL EXPERIENCE

- Data Scientist - Research** | *National Weather Center, University of Oklahoma* | *Norman, Ok* Aug 2024 – Jun 2026
- Analyzed 40+ years of climate data across 100+ meteorological variables to frame a precipitation-forecasting problem, identifying and correcting data quality issues before modeling to improve forecast accuracy by ~15%.
 - Trained and evaluated Random Forest, XGBoost, LSTM, and GRU models with TensorFlow and Scikit-learn on HPC systems, applying cross-validation and statistical hypothesis testing to validate model reliability.
 - Built automated Python and SQL data-wrangling and validation pipelines that cut manual processing effort by ~40%, communicated results to 5+ stakeholders, and contributed modeling work to a co-authored research paper on deep learning-based precipitation forecasting.
- Teaching Assistant - Time Series Analysis** | *University of Oklahoma* | *Norman, Ok* Dec 2025 – May 2026
- Mentored 40+ graduate students on ARIMA modeling, regression, and statistical hypothesis testing using Python, Matlab and R, translating complex time-series methods into applied analytical techniques.
 - Graded and delivered detailed feedback on 100+ assignments and projects, identifying recurring analytical errors and raising cohort proficiency in model evaluation.
- Data Analyst** | *ValueLabs* | *Hyderabad, India* Nov 2023 – Jul 2024
- Wrote complex SQL queries (CTEs, window functions, aggregations, joins) across 120,000+ operational records to clean, validate, and standardize data, improving accuracy by 15%.
 - Designed and deployed 10+ Power BI dashboards with KPI tracking and drill-through filters, cutting manual reporting effort by 25% and accelerating stakeholder decision-making.
 - Partnered with cross-functional stakeholders to translate ambiguous business questions into structured data models and analytical reports supporting operational decisions.

PROJECTS

- Public Health Outbreak Forecasting** | *Python, Time Series, XGBoost, LSTM, Tableau*
- Contributed to the OU-UNSA Public Health and Society Research grant in Arequipa, Peru, applying XGBoost, ARIMA, and LSTM models on 20+ years of public health surveillance data to achieve 0.94 R^2 and a 12% RMSE reduction.
 - Built Tableau dashboards translating outbreak trends for non-technical stakeholders, cutting reporting time by 35% and supporting public health monitoring decisions.
- Enterprise Urban Mobility Data Platform** | *Python, PostgreSQL, FastAPI, Docker, XGBoost, Pytest*
- Designed an end-to-end ETL pipeline with modular transform layers to process 10M+ NYC taxi trip records into PostgreSQL, achieving zero data quality violations across all validation checks.
 - Engineered 8 REST API endpoints via FastAPI serving an XGBoost trip-duration model (R^2 of 0.91), implemented 11 automated Pytest tests, and containerized the platform with Docker.
- Local AI Assistant with RAG Pipeline** | *Python, FastAPI, Ollama, ChromaDB, LLMs*
- Built a fully local retrieval-augmented generation (RAG) pipeline using Ollama-hosted LLMs (Llama 3.2, Phi-4 Mini, Mistral 7B), ChromaDB, and sentence-transformer embeddings, benchmarking 3 models to reach 45 tokens/sec throughput.
 - Designed FastAPI REST endpoints and a Streamlit interface, applying prompt engineering to improve response relevance by ~30% and demonstrate deployable, practical AI tooling.
- National Park Relational Database System** | *SQL, Azure SQL, Java, ER Modeling*
- Designed a normalized (3NF) relational database with ER diagrams and multi-table schemas for a national park management system, reducing data redundancy and enforcing referential integrity across 10+ entities.
 - Implemented SQL stored procedures and triggers in Azure SQL and built a Java front-end application supporting complex multi-table queries and CRUD operations.

EDUCATION

- University of Oklahoma (GPA: 4.0/4.0)** Aug 2024 – May 2026
Master of Science, Data Science and Analytics Norman, OK
- Jawaharlal Nehru Technological University** Dec 2020 – May 2024
Bachelor of Technology, Computer Science and Engineering Hyderabad, India